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CLASS TEST

Q1) Define Reinforcement Learning (RL). Explain its main objectives.

↳ Reinforcement Learning :-

Reinforcement Learning is a branch of machine Learning in which agent learns from interacting with an environment. It performs actions, receives rewards and improve decision through trial and error method.

Objectives:-

- > To learn an optimal policy that maximizes long-term cumulative.
- > learning best strategy through exploration vs exploitation. trade-off

Q2) What is Markov Decision Process (MDP)? List its components?

↳ Markov Decision process is mathematical framework uses model sequential decision-making problems in RL, where next states depends on current state and Action.

Components-

$S \rightarrow$ Set of States

$A \rightarrow$ Set of Actions

$P \rightarrow$ State transition probability

$R \rightarrow$ Reward function

$\gamma \rightarrow$ Discount factor

Q3) State Bellman Equation for State Value Function?

$$\rightarrow V(s) = \sum_a \pi(a|s) \sum_{s', r} P(s', r | s, a) [r + \gamma V(s')]$$

It states that the value of state equals the expected immediate reward plus discounted future value.

It is the fundamental equation used to evaluate policies in RL.

Q4) Differentiate between Exploration and Exploitation?

$\rightarrow$ Exploration:- The agent tries new or less known actions to gather more information about the environment.

Exploitation:- The agent selects the action with highest known reward to maximize performance.

Exploration improves learning, while exploitation maximizes immediate reward.

Q5) What is Multi-Armed Bandit Problem?

$\rightarrow$ Multi-Armed Bandit Problem is a reinforcement learning problem in which an agent must choose among several actions (slot machine), each with an unknown reward.

The goal is to maximize total reward by balancing exploration and exploitation.

Q6) What is Dynamic Programming (DP)? Mention its Applications in RL?

$\rightarrow$ Dynamic Programming is a method for solving

complex problems by breaking them into smaller subproblems and using previous computed results.

Applications:-

- Policy Evaluation
- Policy Improvement
- Policy Iteration
- Value Iteration.

Q7> What is Temporal Difference (TD) learning? write TD(0) update Equation?

↳ Temporal Difference (TD) learning:- It is a RL method that updates value estimate using difference between predicted and actual reward with waiting for final outcome.

TD(0) Update Equation:-

$$V(s) \leftarrow V(s) + \alpha [r + \gamma V(s') - V(s)]$$

α - learning rate

γ - Discount factor

r - Immediate Reward.

Q8> Differentiate between SARSA & Q-learning.

SARSA	Q-learning
→ ON-policy → Updates Q-value using next action actually taken → Learning policy being followed → more conservative and safer	→ OFF-policy → Updates Q-value using maximum possible next Q-value. → learns optimal policy regardless of current policy → learns faster

Q9 What is Deep-Q Network (DQN)? Mention its advantages over tabular Q-learning?

↳ Deep-Q Network (DQN):- DQN uses a deep neural network that uses Q-table (Q-learning) to estimate Q-values instead of Q-table.

Advantages:-

- Handles large state space
- Generalized to new state space
- No need of Q-table
- Estimates using Q-Values

Q10 What is Double Deep Q-Network (DDQN)? Why is it preferred over DQN?

↳ Double Deep Q-Network (DDQN):- It is an improved version of (DQN) uses two networks to reduce overestimations of Q-values.

Why it is preferred?

- Reduces Overestimation
- More stable learning
- Better performance than DQN.